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Team 24 CPEN 333 Final Project

By: Cole Shanks, Sandrine Yip, Karmen Wang, Jordan Lee

Requirements Satisfied

Central Computer

- Robot movement with collision avoidance
- The ability to dock and charge robots
- Manage inventory and item locations for all products
- Coordinate the arrival and departure of Delivery and Stocking Trucks

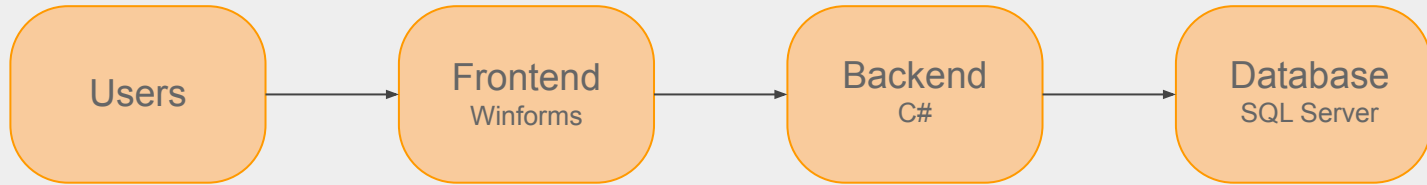
User Interface

- Separate logins and interfaces for Customers and Managers
- Check the number of stock for an item
- Receive alerts for out-of-stock items
- Dynamically add warehouses of variable dimensions (Manager Mode)

BONUS

- Create new user accounts with unique login credentials
- Permanently remove discontinued items from the database
- Add new items to the database and stock them in the warehouse
- Updating list that keeps track of number of warehouses, items, users, and revenue

Tech Stack



Front-end (Winforms)

- Easy learning curve (We are familiar with C#)
- Has event handling and pre-made components

Customers

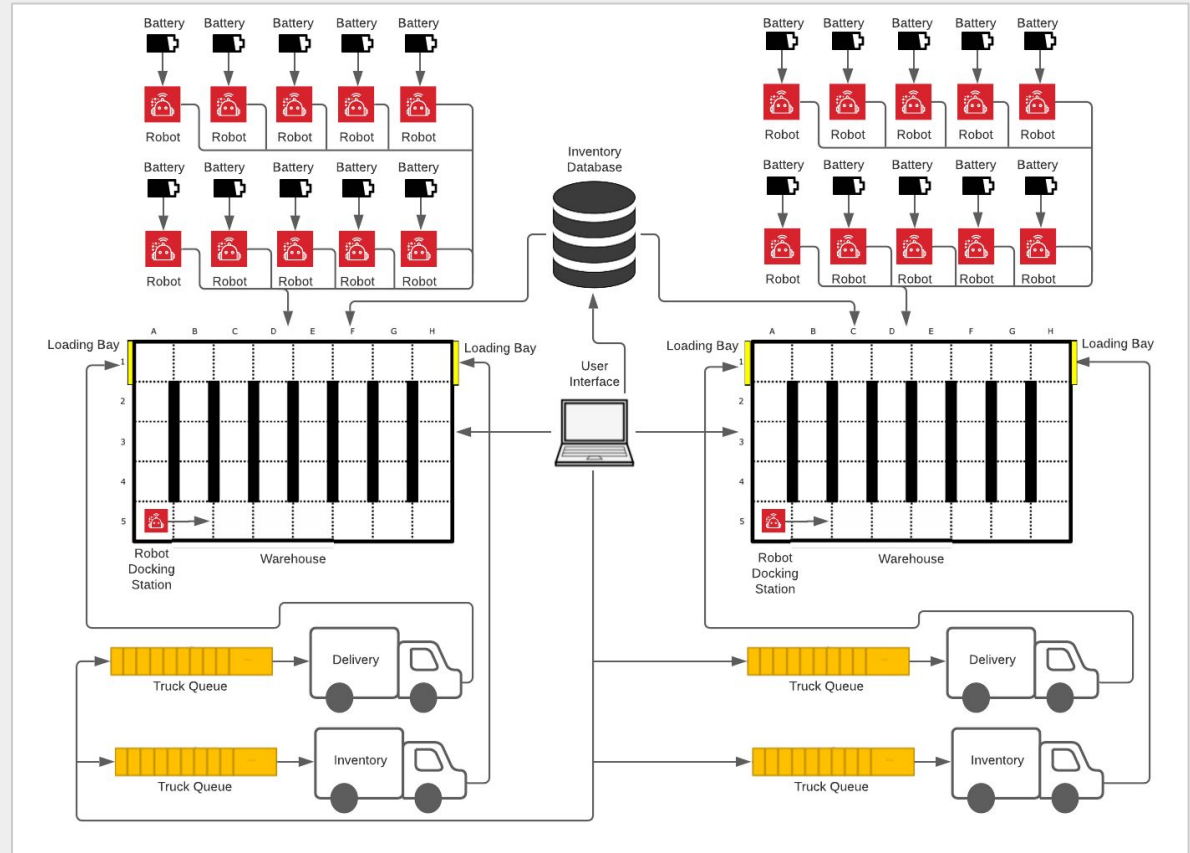
- Can purchase items
- Check item availability
- Filter search catalog

Manager

- Add Item to inventory
- Add Warehouses
- View general information
- Restock Item
- Send delivery trucks

Back-end

- Mutexed the dictionary
 - Read and write
- List of integers
- Concurrent queue for trucks
 - FIFO
- Concurrent stack for robots
 - LIFO
- Database vs in line code



Purchase Items Use Case

Start of Primary scenario/transaction

1. **Include** Customer Login
2. If Customer chooses to logout << Scenario 1 >>
3. If Customer selects Check Item Availability << Scenario 2 >>
4. Customer selects items from the catalog on the left by checking boxes
5. Customer selects Add to Cart
6. If the Item is out of stock
Extend Use Case: Customer is informed that the item is out of stock
7. If the Customer selects Remove from Cart << Scenario 3 >>
8. The Customer selects Buy Items
9. The central computer informs the Customer that the order has been placed
10. The central computer displays robot movement information in the terminal

End-Of-Transaction

Scenario 1: Return Customer to the Sign in screen. End of Transaction

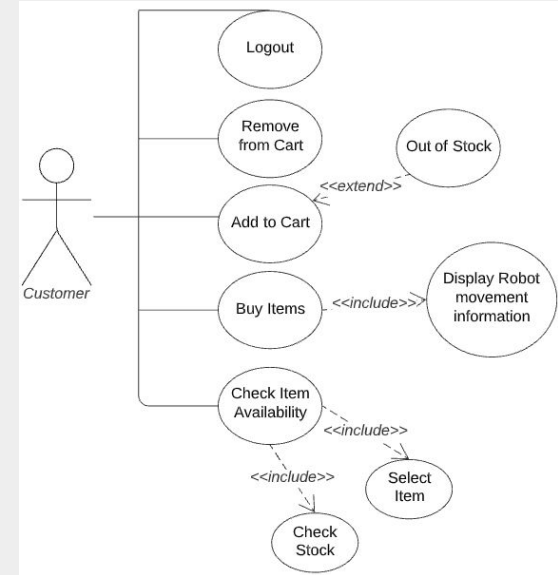
Scenario 2: Customer selects an Item from the dropdown and presses check stock

The central computer checks the item's stock in the database and informs the Customer

Resume primary scenario

Scenario 3: The central computer removes this item from the cart

Resume primary scenario



Demo Time!